

SPECIAL SUMMER INTEGRATED LEARNING PROGRAMME MATHEMATICS

Introduction to Integral Calculus

Week 6 Day 2 AM



SINGAPORE UNIVERSITY OF
TECHNOLOGY AND DESIGN

OUTLINE

① The Fundamental Theorem of Calculus

② Integration by Substitution

③ Summary

TWO DISTINCT MAIN CONCEPTS

Note that a priori, indefinite integrals and definite integrals are two distinct concepts that are defined differently.

- **Indefinite integrals:** Reverse operation of differentiation. If $F'(x) = f(x)$, then

$$\int f(x) \, dx = F(x) + c$$

- **Definite integrals:**

$$\int_a^b f(x) \, dx = \text{Signed area of the region between the graph of } f(x)$$

and the x -axis from $x = a$ to $x = b$.

The two distinct concepts are connected together by the Fundamental Theorem of Calculus (FTC), which is probably the most important theorem in calculus (there are two versions, together they connect differentiation, indefinite integrals and definite integrals). The proof is a bit technical and non-trivial, which we'll omit.

THE FUNDAMENTAL THEOREM OF CALCULUS

THE 2ND FUNDAMENTAL THEOREM OF CALCULUS

Let $f(x)$ be a **continuous** function defined on a finite closed interval $[a, b]$. Suppose we know an antiderivative $F(x)$ of $f(x)$, i.e.

$$F'(x) = f(x),$$

then

$$\int_a^b f(x) \, dx = F(b) - F(a).$$

Notation:

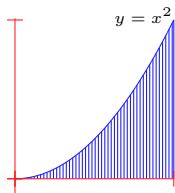
$$F(x) \Big|_a^b := F(b) - F(a).$$

In layman terms, the theorem says that the definite integral of **continuous** $f(x)$ over the interval $[a, b]$ is equal to the difference of **any** antiderivative of $f(x)$ evaluated at $x = b$ and $x = a$.

2ND FTC - EXAMPLE

E.g.: Suppose we want to find the area under the curve $y = x^2$ from $x = 0$ to $x = 1$.

An antiderivative for x^2 would be $\frac{1}{3}x^3$. So the 2nd FTC implies



$$\int_0^1 x^2 dx = \left. \frac{1}{3}x^3 \right|_0^1 = \frac{1}{3} \cdot 1^3 - \frac{1}{3} \cdot 0^3 = \frac{1}{3}.$$

This is what we found in the last class by taking limits.

It is easier to compute definite integrals using antiderivatives than using limits (this was discovered by Newton, Leibniz and others), provided we can find at least one of them! This leads us to our next topic: integration techniques, i.e. methods to find antiderivatives.

2ND FTC - EXAMPLE

E.g.: A bungee jumper leaps off the starting platform at time $t = 0$ s and rebounds once during the first 5 seconds. With velocity measured downward, for t in seconds and $0 \leq t \leq 5$, the jumper's velocity is approximated by $v(t) = -4t^2 + 16t \text{ ms}^{-1}$. Then

$\int_0^5 v(t) dt = s(5) - s(0)$ is the displacement of the jumper relative to the starting point, which is

$$\begin{aligned} s(5) - s(0) &= \int_0^5 -4t^2 + 16t dt \\ &= \left(-\frac{4}{3}t^3 + 8t^2 \right) \Big|_0^5 \\ &= \frac{100}{3} \text{ m.} \end{aligned}$$

ACTIVITY 1

Compute following definite integrals whenever the FTC is applicable.

(a) $\int_0^1 5 \, dx.$

(b) $\int_{-1}^2 t \, dt.$

(c) $\int_1^4 \sqrt{z} \, dz.$

(d) $\int_{-1}^1 \frac{1}{t} \, dt.$

(e) $\int_{-2}^{-1} \frac{1}{t^3} \, dt.$

(f) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x \, dx.$

ACTIVITY 1: SOLUTION

$$(a) \int_0^1 5 \, dx = 5x \Big|_0^1 = 5(1) - 5(0) = 5.$$

$$(b) \int_{-1}^2 t \, dt = \frac{1}{2}t^2 \Big|_{-1}^2 = \frac{1}{2}(2^2) - \frac{1}{2}(-1)^2 = \frac{3}{2}.$$

$$(c) \int_1^4 \sqrt{z} \, dz = \frac{2}{3}z^{\frac{3}{2}} \Big|_1^4 = \frac{2}{3}4^{\frac{3}{2}} - \frac{2}{3} = \frac{14}{3}.$$

(d) The FTC is not applicable as $\frac{1}{t}$ is not continuous over $[-1, 1]$.

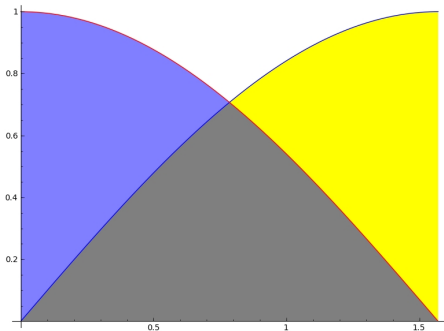
$$(e) \int_{-2}^{-1} \frac{1}{t^3} \, dt = \left(-\frac{1}{2t^2} \right) \Big|_{-2}^{-1} = -\frac{1}{2(-1)^2} + \frac{1}{2(-2)^2} = -\frac{3}{8}.$$

(f) Since $\cos x$ is an even function,

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x \, dx = 2 \int_0^{\frac{\pi}{2}} \cos x \, dx = 2 \sin x \Big|_0^{\frac{\pi}{2}} = 2 \left(\sin \frac{\pi}{2} - \sin 0 \right) = 2.$$

ACTIVITY 2

Find the area of the region bounded between the curves $y = \sin x$ and $y = \cos x$ over the interval $0 \leq x \leq \frac{\pi}{2}$.



Hint: The region mentioned is the blue and yellow regions.

ACTIVITY 2: SOLUTION

By solving $\sin x = \cos x$, the two functions intersect at $x = \frac{\pi}{4}$ in the interval $0 \leq x \leq \frac{\pi}{2}$.

Using symmetry, the area of the blue region is equal to the area of the yellow region. So the required area is

$$\begin{aligned}
 A &= 2 \left(\int_0^{\frac{\pi}{4}} \cos x \, dx - \int_0^{\frac{\pi}{4}} \sin x \, dx \right) \\
 &= 2 \left(\sin x \Big|_0^{\frac{\pi}{4}} + \cos x \Big|_0^{\frac{\pi}{4}} \right) \\
 &= 2 \left(\frac{\sqrt{2}}{2} - 0 + \frac{\sqrt{2}}{2} - 1 \right) \\
 &= 2(\sqrt{2} - 1).
 \end{aligned}$$

OUTLINE

① The Fundamental Theorem of Calculus

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INTEGRATION BY SUBSTITUTION

The chain rule

$$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x),$$

implies the formula of **integration by substitution**:

INTEGRATION BY SUBSTITUTION

$$\int f'(g(x))g'(x) dx = f(g(x)) + C,$$

which can be interpreted as substituting $u = g(x)$ and so
 $\frac{du}{dx} = g'(x) \Rightarrow 'du = g'(x)dx'$

$$\begin{aligned}\int f'(g(x))g'(x)dx &= \int f'(u) du \\ &= f(u) + C \\ &= f(g(x)) + C.\end{aligned}$$

INTEGRATION BY SUBSTITUTION - CASE 1

Case 1: If the pair of $g(x)$ and $g'(x)$ appear clearly (up to a coefficient) in the integrand, we should try to substitute $u = g(x)$.

E.g.: Find $\int \sqrt{x^3 + 2} x^2 dx$.

Since $\frac{d}{dx}(x^3 + 2) = 3x^2$ appears (up to a coefficient) in the integrand, it suggests that we should substitute $u = x^3 + 2$ and $\frac{du}{dx} = 3x^2$,

$$\begin{aligned}\int \sqrt{x^3 + 2} x^2 dx &= \int \left(\frac{\sqrt{x^3 + 2}}{3} \right) 3x^2 dx \\ &= \int \frac{\sqrt{u}}{3} du = \frac{2u^{\frac{3}{2}}}{9} + C.\end{aligned}$$

Back substituting,

$$\int \sqrt{x^3 + 2} x^2 dx = \frac{2u^{\frac{3}{2}}}{9} + C = \frac{2}{9}(x^3 + 2)^{\frac{3}{2}} + C.$$

INTEGRATION BY SUBSTITUTION - CASE 2

Case 2: $g(x)$ is hidden but the choice of $g'(x)$ is obvious. So we find $u = g(x)$ as the antiderivative of $g'(x)$.

E.g.: Find $\int \sin^3 x \, dx$.

There are 3 possible choices (up to a coefficient) for $g'(x)$, which are $\sin^3 x$, $\sin^2 x$ or $\sin x$. However, if $g'(x)$ is $\sin^3 x$ or $\sin^2 x$, it is hard to find its antiderivative $g(x)$. So we are left with the option $g'(x) = \sin x$ and hence $g(x) = -\cos x$. This suggests us to manipulate the integrand in the following way:

$$\begin{aligned}\int \sin^3 x \, dx &= \int \sin^2 x \sin x \, dx \\ &= \int (1 - \cos^2 x) \sin x \, dx.\end{aligned}$$

INTEGRATION BY SUBSTITUTION - CASE 2

$$\begin{aligned}\int \sin^3 x \, dx &= \int \sin^2 x \sin x \, dx \\ &= \int (1 - \cos^2 x) \sin x \, dx.\end{aligned}$$

Now, let $u = \cos x$ and hence $\frac{du}{dx} = -\sin x$, we get

$$\begin{aligned}\int \sin^3 x \, dx &= \int (1 - u^2)(-1) \, du \\ &= \int (u^2 - 1) \, du \\ &= \frac{1}{3}u^3 - u + C \\ &= \frac{1}{3}\cos^3 x - \cos x + C.\end{aligned}$$

INTEGRATION BY SUBSTITUTION - CASE 3

Case 3: The choice of $g(x)$ and $g'(x)$ are not obvious. Try substituting $u = g(x)$ as one of the complicated expressions in the integrand.

E.g.: Find $\int \frac{\sqrt{x}}{\sqrt{x} + 1} dx$.

We try letting $u = \sqrt{x} + 1$ and so $\frac{du}{dx} = \frac{1}{2\sqrt{x}}$. Although $\frac{1}{2\sqrt{x}}$ doesn't appear as a factor in the integrand, we can always introduce it by multiplying $\frac{2\sqrt{x}}{2\sqrt{x}}$:

$$\begin{aligned}\int \frac{\sqrt{x}}{\sqrt{x} + 1} dx &= \int \frac{\sqrt{x}}{\sqrt{x} + 1} \frac{2\sqrt{x}}{2\sqrt{x}} dx \\ &= \int \frac{2x}{\sqrt{x} + 1} \frac{1}{2\sqrt{x}} dx,\end{aligned}$$

and we substitute $u = \sqrt{x} + 1 \Rightarrow x = (u - 1)^2$ and $\frac{du}{dx} = \frac{1}{2\sqrt{x}}$.

INTEGRATION BY SUBSTITUTION - CASE 3

$$u = \sqrt{x} + 1 \Rightarrow x = (u - 1)^2 \text{ and } \frac{du}{dx} = \frac{1}{2\sqrt{x}}.$$

$$\begin{aligned} \int \frac{\sqrt{x}}{\sqrt{x} + 1} dx &= \int \frac{2x}{\sqrt{x} + 1} \frac{1}{2\sqrt{x}} dx \\ &= \int \frac{2(u - 1)^2}{u} du \\ &= 2 \int \frac{u^2 - 2u + 1}{u} du \\ &= 2 \int u - 2 + \frac{1}{u} du \\ &= 2 \left(\frac{1}{2} u^2 - 2u + \ln |u| \right) + C \\ &= (\sqrt{x} + 1)^2 - 4(\sqrt{x} + 1) + 2 \ln(\sqrt{x} + 1) + C. \end{aligned}$$

REMARKS ON INTEGRATION BY SUBSTITUTION

In the previous example, where we made the substitution $u = \sqrt{x} + 1$, we don't actually have to multiply $\frac{2\sqrt{x}}{2\sqrt{x}}$ explicitly in order to get $\frac{du}{dx} = \frac{1}{2\sqrt{x}}$, this step is equivalent to substituting

$$\frac{du}{dx} = \frac{1}{2\sqrt{x}}$$

$$\Rightarrow 2\sqrt{x} \, du = dx$$

$$\Rightarrow dx = 2\sqrt{x} \, du = 2(u - 1) \, du$$

into the original integration:

$$\begin{aligned} \int \frac{\sqrt{x}}{\sqrt{x} + 1} dx &= \int \frac{u - 1}{u} 2(u - 1) \, du \\ &= \int \frac{2(u - 1)^2}{u} \, du = \dots \\ &= (\sqrt{x} + 1)^2 - 4(\sqrt{x} + 1) + 2 \ln(\sqrt{x} + 1) + C. \end{aligned}$$

REMARKS ON INTEGRATION BY SUBSTITUTION

In general, once we choose the substitution $u = g(x)$, we can just substitute $du = g'(x) dx$ or $dx = \frac{1}{g'(x)} du$ into the integrand.

This simplifies the formula

$$\int f'(g(x))g'(x) dx = f(g(x)) + C$$

into

$$\int f'(u)du = f(u) + C.$$

Remember that $\frac{du}{dx}$ is an atomic notation which means the derivative of u with respect to x and we cannot treat $\frac{du}{dx}$ as a fraction.

However, writing $du = g'(x) dx$ or $dx = \frac{1}{g'(x)} du$ provides us a convenient way to apply integration by substitution. You can view this as a notational mnemonic for the integration by substitution formula.

INTEGRATION BY SUBSTITUTION FOR DEFINITE INTEGRALS

Suppose we have $f(x) = F'(x)$.

- From the Fundamental Theorem of Calculus and integration by substitution:

$$\begin{aligned}\int_a^b f(g(x)) g'(x) dx &= \int_a^b F'(g(x)) g'(x) dx \\ &= F(g(x)) \Big|_a^b = F(g(b)) - F(g(a)).\end{aligned}$$

- but also with the substitution $u = g(x)$,

$$\int_{g(a)}^{g(b)} f(u) du = F(g(b)) - F(g(a)).$$

Therefore:

$$\int_a^b f(g(x)) g'(x) dx = \int_{g(a)}^{g(b)} f(u) du.$$

INTEGRATION BY SUBSTITUTION FOR DEFINITE INTEGRALS - EXAMPLE

E.g.: Compute the integral $\int_e^{e^2} \frac{1}{x \ln x} dx$ in **two** ways.

1. We find the **indefinite** integral $\int \frac{1}{x \ln x} dx$ first.

- Choose $u = \ln x$ and hence $\frac{du}{dx} = \frac{1}{x}$, we get

$$\int \frac{1}{x \ln x} dx = \int \frac{1}{\ln x} \cdot \frac{1}{x} dx = \int \frac{1}{u} du = \ln |u| + C.$$

- Back substituting

$$\int \frac{1}{x \ln x} dx = \ln |\ln x| + C.$$

- Hence

$$\begin{aligned} \int_e^{e^2} \frac{1}{x \ln x} dx &= \ln |\ln x| \Big|_e^{e^2} = \ln (\ln e^2) - \ln (\ln e) \\ &= \ln 2 - \ln 1 = \ln 2. \end{aligned}$$

INTEGRATION BY SUBSTITUTION FOR DEFINITE INTEGRALS - EXAMPLE

E.g.: Compute the integral $\int_e^{e^2} \frac{1}{x \ln x} dx$ in **two** ways.

2. Substitute the lower and upper limits in the definite integral.

- Choose $u = g(x) = \ln x$ and hence $\frac{du}{dx} = \frac{1}{x}$.
- Substitute the limits of integration:
 $a = e \Rightarrow g(a) = \ln e = 1, b = e^2 \Rightarrow g(b) = \ln e^2 = 2.$
- Compute the value

$$\int_e^{e^2} \frac{1}{x \ln x} dx = \int_1^2 \frac{1}{u} du = \ln |u| \Big|_1^2 = \ln 2 - \ln 1 = \ln 2.$$

Summary: If you express the antiderivative in terms of x , you should use the lower and upper limits a, b of x . If you express the antiderivative in terms of $u = g(x)$, you should use the lower and upper limits $g(a), g(b)$ of u .

ACTIVITY 3

Solve the following integrals.

(a) $\int_0^1 \frac{x}{1+x^2} dx.$

(b) $\int_0^{\frac{\pi}{4}} \tan x dx.$

(c) $\int \cos(3x-4) dx.$

(d) $\int e^{-3x} dx.$

(e) $\int \frac{1}{\sqrt{4-x}} dx.$

(f) $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx.$

(g) $\int \frac{1}{x^2+a^2} dx,$ for any non-zero constant a .

ACTIVITY 3: SOLUTION

- (a) Since $\frac{d}{dx}(x^2 + 1) = 2x$ appears (up to a coefficient) in the integrand, it suggests that we should substitute $u = x^2 + 1$ and $\frac{du}{dx} = 2x \Leftrightarrow x \, dx = \frac{1}{2} \, du$. Also, $u = 0^2 + 1 = 1$ when $x = 0$, $u = 1^2 + 1 = 2$ when $x = 1$.

$$\begin{aligned}\int_0^1 \frac{x}{1+x^2} \, dx &= \int_1^2 \frac{1}{u} \cdot \frac{1}{2} \, du \\ &= \frac{1}{2} \ln |u| \Big|_1^2 \\ &= \frac{1}{2} (\ln 2 - \ln 1) = \frac{1}{2} \ln 2.\end{aligned}$$

ACTIVITY 3: SOLUTION

- (b) Since $\tan x = \frac{\sin x}{\cos x}$ and $\frac{d}{dx}(\cos x) = -\sin x$ appears (up to a coefficient) in the integrand, it suggests that we should substitute $u = \cos x$ and
- $$\frac{du}{dx} = -\sin x \Leftrightarrow \sin x \, dx = (-1) \, du. \text{ Also, } u = \cos 0 = 1$$
- when $x = 0$, $u = \cos \frac{\pi}{4} = \frac{\sqrt{2}}{2}$ when $x = \frac{\pi}{4}$.

$$\begin{aligned}\int_0^{\frac{\pi}{4}} \tan x \, dx &= \int_0^{\frac{\pi}{4}} \frac{\sin x}{\cos x} \, dx \\&= \int_1^{\frac{\sqrt{2}}{2}} \frac{1}{u}(-1) \, du = \int_{\frac{\sqrt{2}}{2}}^1 \frac{1}{u} \, du \\&= \ln |u| \Big|_{\frac{\sqrt{2}}{2}}^1 = \ln 1 - \ln \frac{\sqrt{2}}{2} \\&= 0 - \ln \frac{1}{\sqrt{2}} = \ln \sqrt{2}.\end{aligned}$$

ACTIVITY 3: SOLUTION

- (c) Since $\frac{d}{dx}(3x - 4) = 3$ appears (up to a coefficient) in the integrand, it suggests that we should substitute $u = 3x - 4$ and $\frac{du}{dx} = 3 \Leftrightarrow dx = \frac{1}{3} du$.

$$\int \cos(3x - 4) dx = \int \cos u \frac{1}{3} du = \frac{1}{3} \sin u + C = \frac{1}{3} \sin(3x - 4) + C.$$

Remark: In general, for any constants $a \neq 0$, $b \in \mathbb{R}$, we have

$$\int f'(ax + b) dx = \frac{1}{a} f(ax + b) + C.$$

This only works for linear function $ax + b$. It doesn't work if it was non-linear, for instance

$$\int f'(ax^2 + b) dx \neq \frac{1}{2ax} f(ax^2 + b) + C,$$

because $2ax$ is not a constant. When we differentiate the right-hand-side, we need the quotient rule and it won't give the integrand on the left-hand-side.

ACTIVITY 3: SOLUTION

- (d) Since $\frac{d}{dx}(-3x) = -3$ appears (up to a coefficient) in the integrand, it suggests that we should substitute $u = -3x$ and $\frac{du}{dx} = -3 \Leftrightarrow dx = -\frac{1}{3} du$.

$$\begin{aligned}\int e^{-3x} dx &= \int e^u \left(-\frac{1}{3}\right) du \\ &= -\frac{1}{3}e^u + C = -\frac{1}{3}e^{-3x} + C.\end{aligned}$$

- (e) We substitute $u = 4 - x$ and $\frac{du}{dx} = -1 \Leftrightarrow dx = (-1) du$.

$$\begin{aligned}\int \frac{1}{\sqrt{4-x}} dx &= \int \frac{1}{\sqrt{u}}(-1) du = -\int u^{-\frac{1}{2}} du \\ &= -\left(\frac{1}{\frac{1}{2}}u^{\frac{1}{2}}\right) + C \\ &= -2(4-x)^{\frac{1}{2}} + C = -2\sqrt{4-x} + C.\end{aligned}$$

ACTIVITY 3: SOLUTION

(f) Since $\frac{d}{dx} \sqrt{x} = \frac{1}{2\sqrt{x}}$ appears (up to a coefficient) in the integrand, it suggests that we should substitute $u = \sqrt{x}$ and $\frac{du}{dx} = \frac{1}{2\sqrt{x}} \Leftrightarrow \frac{1}{\sqrt{x}} dx = 2 du$.

$$\begin{aligned} \int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx &= \int e^u \cdot 2 du \\ &= 2e^u + C \\ &= 2e^{\sqrt{x}} + C. \end{aligned}$$

ACTIVITY 3: SOLUTION

(g) Since $\int \frac{1}{x^2 + a^2} dx = \frac{1}{a^2} \int \frac{1}{(\frac{x}{a})^2 + 1} dx$, we can substitute $u = \frac{x}{a} \Rightarrow \frac{du}{dx} = \frac{1}{a} \Rightarrow dx = a du$. So

$$\begin{aligned} \int \frac{1}{x^2 + a^2} dx &= \frac{1}{a^2} \int \frac{1}{(\frac{x}{a})^2 + 1} dx \\ &= \frac{1}{a^2} \int \frac{1}{u^2 + 1} \cdot a du \\ &= \frac{1}{a} \arctan u + C = \frac{1}{a} \arctan \frac{x}{a} + C. \end{aligned}$$

⚠ Note that it's possible to get different expressions for the same indefinite integral, as antiderivatives are not unique and the constants might not be the same. The best way to check your answer is to differentiate it and you should get back the integrand.

ACTIVITY 4

Solve the following integrals.

(a) $\int x\sqrt{x+1} \, dx.$

(b) $\int \frac{1}{1+\sqrt{x}} \, dx.$

(c) $\int \frac{x}{\sqrt{1+x}} \, dx.$

(d) $\int (x+2)\sqrt{1-x} \, dx.$

(e) $\int \cos^3 x \, dx.$

(f) $\int \sec x \, dx.$

Hint: multiply by $\frac{\sec x + \tan x}{\sec x + \tan x}.$

ACTIVITY 4: SOLUTION

- (a) We substitute $u = \sqrt{x+1}$, which is one of the complicated expressions in the integrand. So $x = u^2 - 1$ and $\frac{dx}{du} = 2u \Leftrightarrow dx = 2u \, du$.

$$\begin{aligned}\int x\sqrt{x+1} \, dx &= \int (u^2 - 1) \cdot u \cdot 2u \, du \\&= \int (2u^4 - 2u^2) \, du \\&= \frac{2}{5}u^5 - \frac{2}{3}u^3 + C \\&= \frac{2}{5}(x+1)^{\frac{5}{2}} - \frac{2}{3}(x+1)^{\frac{3}{2}} + C.\end{aligned}$$

ACTIVITY 4: SOLUTION

- (b) We substitute $u = 1 + \sqrt{x}$, which is one of the complicated expressions in the integrand. So $x = (u - 1)^2$ and $\frac{dx}{du} = 2(u - 1) \Leftrightarrow dx = 2(u - 1) du$.

$$\begin{aligned}\int \frac{1}{1 + \sqrt{x}} dx &= \int \frac{1}{u} \cdot 2(u - 1) du \\ &= \int \left(2 - \frac{2}{u} \right) du \\ &= 2u - 2 \ln |u| + C \\ &= 2(1 + \sqrt{x}) - 2 \ln(1 + \sqrt{x}) + C.\end{aligned}$$

ACTIVITY 4: SOLUTION

- (c) We substitute $u = \sqrt{1+x}$, which is one of the complicated expressions in the integrand. So $x = u^2 - 1$ and $\frac{dx}{du} = 2u \Leftrightarrow dx = 2u \, du$.

$$\begin{aligned}\int \frac{x}{\sqrt{1+x}} \, dx &= \int \frac{u^2 - 1}{u} \cdot 2u \, du \\&= \int (2u^2 - 2) \, du \\&= \frac{2}{3}u^3 - 2u + C \\&= \frac{2}{3}(1+x)^{\frac{3}{2}} - 2\sqrt{1+x} + C.\end{aligned}$$

ACTIVITY 4: SOLUTION

- (d) We substitute $u = \sqrt{1-x}$, which is one of the complicated expressions in the integrand. So $x = 1 - u^2$ and $\frac{dx}{du} = -2u \Leftrightarrow dx = -2u \, du$.

$$\begin{aligned}\int (x+2)\sqrt{1-x} \, dx &= \int (1-u^2+2) \cdot u \cdot (-2u) \, du \\ &= \int (2u^4 - 6u^2) \, du \\ &= \frac{2}{5}u^5 - 2u^3 + C \\ &= \frac{2}{5}(1-x)^{\frac{5}{2}} - 2(1-x)^{\frac{3}{2}} + C.\end{aligned}$$

ACTIVITY 4: SOLUTION

(e) Since $\cos^3 x = \cos^2 x \cos x = (1 - \sin^2 x) \cos x$ and $\frac{d}{dx} \sin x = \cos x$, we substitute $u = \sin x$. So $\frac{du}{dx} = \cos x \Leftrightarrow \cos x \, dx = du$.

$$\begin{aligned}\int \cos^3 x \, dx &= \int (1 - \sin^2 x) \cos x \, dx \\&= \int (1 - u^2) \, du \\&= u - \frac{1}{3}u^3 + C \\&= \sin x - \frac{1}{3}\sin^3 x + C.\end{aligned}$$

ACTIVITY 4: SOLUTION

(f) After multiplying the integrand by $1 = \frac{\sec x + \tan x}{\sec x + \tan x}$, we get

$$\int \sec x \, dx = \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} \, dx.$$

Since $\frac{d}{dx}(\sec x + \tan x) = \sec x \tan x + \sec^2 x$, we substitute $u = \sec x + \tan x$. So

$$\frac{du}{dx} = \sec x \tan x + \sec^2 x \Leftrightarrow (\sec x \tan x + \sec^2 x) \, dx = du.$$

$$\begin{aligned} \int \sec x \, dx &= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} \, dx \\ &= \int \frac{1}{u} \, du \\ &= \ln |u| + C \\ &= \ln |\sec x + \tan x| + C. \end{aligned}$$

OUTLINE

① The Fundamental Theorem of Calculus

② Integration by Substitution

③ Summary

SUMMARY

- The Fundamental Theorem of Calculus
 - the 2nd Fundamental Theorem of Calculus.
 - computing definite integrals by antiderivatives.
- Integration by Substitution
 - the chain rule and integration by substitution.
 - cases to apply integration by substitution.
 - integration by substitution for definite integrals.

RESOURCES

References:

- George F. Simmons. Calculus with Analytic Geometry (2nd edition) sections 6.6, 5.3.
- Hughes-Hallett. Calculus (6th edition) sections 6.4, 7.1.